George Panagopoulos

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Experience

- **Amazon**, Applied Scientist Intern, Luxembourg (04.2022 - Now)
  - Machine Learning for Operations Research
- **Ecole Polytechnique Informatics Laboratory (LIX)**, Research Engineer, Paris, France (06.2018 - 03.2022)
  - **Graph and reinforcement learning** for combinatorial optimization and epidemic forecasting
  - **Data mining** from big bibliographical networks, legal documents, medical ontologies
- **University of Houston, Computational Physiology Lab**, RA/TA, Houston, TX (08.2016 - 05.2018)
  - **Forecasting** driving anomalies using physiological signals from wearable sensors
  - **Statistical analysis and inference** on behavioral data
  - TA Courses: Statistical Methods in Research & Software Engineering
- **National Center for Scientific Research Demokritos**, Research Assistant, Athens, Greece (08.2014 - 07.2016)
  - **Brain computer interface** tutorials for university students and high-school students
  - **Java web services** modeling computational creativity in multi-lingual essays

Education

- **Ph.D. in Computer Science**, Ecole Polytechnique, Paris, France (01.2019 - 03.2022)
  - Thesis Title: *"Learning Influence Representations: Methods and Applications"*
  - Advisor: Prof. Michalis Vazirgiannis
  - Co-advisor: Asst Prof. Fragkiskos Malliaros
- **Ms. in Computer Science**, University of Houston, Houston, TX (08.2016 - 05.2018)
  - Thesis Title: *"Forecasting Markers of Habitual Driving Behaviors Associated with Crash Risk"*
  - Advisor: Prof. Ioannis Pavlidis
- **Bsc. in Informatics and Telematics**, Harokopio University of Athens, Greece (09.2010 - 07.2014)
  - Thesis Title: *"Detecting Rising Stars in Dynamic Collaborative Networks"*
  - Advisor: Prof. Iraklis Varlamis

Selected Publications

1. G. Panagopoulos, N. Tziortziotis, F. Malliaros, M. Vazirgiannis. *"Learning Graph Representations for Influence Maximization"*.
2. G. Nikolentzos, G. Panagopoulos, M. Vazirgiannis. *"An Empirical Study of the Expressiveness of Graph Kernels and Graph Neural Networks"*, in International Conference on Artificial Neural Networks (**ICANN**), (2021).
3. B. Rozemberczki , P. Scherer, Y. He, G. Panagopoulos, M. Astefanoaei, O. Kiss, F. Beres, N. Collignon, R. Sarkar. *"PyTorch Geometric Temporal: Spatiotemporal Signal Processing with Neural Machine Learning Models"*. in Conference on Information and Knowledge Management (**CIKM resource track**) , 2021.
4. G. Panagopoulos, G. Nikolentzos, M. Vazirgiannis.*"Transfer Graph Neural Networks for Pandemic Forecasting"*, in AAAI Conference on Artificial intelligence (**AAAI**), 2021.
5. G. Panagopoulos, F. Malliaros, M. Vazirgiannis. *"Multi-task Learning for Influence Estimation and Maximization"*, in **IEEE Transactions of Knowledge and Data Engineering**, 2020, 10.1109/TKDE.2020.3040028.
6. G. Panagopoulos, H. Jalalzai *"Graph Neural Networks with Extreme Nodes Discrimination"*, in Deep Learning on Graphs: Methods and Applications, (**DLG@KDD**) , 2020
7. G. Panagopoulos, F. Malliaros, M. Vazirgiannis. *"Influence Maximization using Influence and Susceptibility Embeddings"*, in AAAI International Conference on Web and Social Media (**ICWSM**), 2020
8. G. Panagopoulos, I. Pavlidis. *"Forecasting Markers of Habitual Driving Behaviors Associated with Crash Risk"* **IEEE Transactions of Intelligent Transportation Systems** (2019)
9. I. Pavlidis, D. Zavlin, A. Khatri, A. Wesley, G. Panagopoulos, and A. Echo. *"Absence of Stressful Conditions Accelerates Dexterous Skill Acquisition in Surgery"* **Nature Scientific Reports** 9.1 (2019):1747
10. G. Panagopoulos, G. Tsatsaronis, and I. Varlamis. *"Detecting Rising Stars in Dynamic Collaborative Networks"* **Elsevier Journal of Informetrics** 11.1 (2017): 198-222.

Awards & Scholarships

- KDD Registration Award, 2020
- ICWSM Best paper honorable mention, 2020
- Best Ms Thesis Award, University of Houston, 2018
- PETRA Travel Award, 2017
- Presidential Fellowship Award, University of Houston, 2016
- PhD Fellowship, University of Houston, 2016
- Doctoral Scholarship, NCSR Demokritos, 2016
- 1st place in Open Public Data Hackathon, Greek Ministry of Administrative Reform, 2014
- 1st place in Green Hackathon, Agro-Know, 2013

Technical Skills

- Coding: Python (advanced), Java (intermediate), C++ (basic)
- Statistics and plots: R (advanced), Matlab, Julia (basic)
- Deep Learning: PyTorch (advanced), Tensorflow (intermediate)
- Databases: MySQL, PostgreSQL (advanced), Spark, Flink, Hadoop (basic)
- Optimization : CVX, CVXOPT (basic)

Scientific Services

- Reviewer: Knowledge Discovery and Data Mining (KDD 2020,2019), International Conference on Machine Learning (ICML 2019), The Web Conference (WWW 2019), International Conference on Artificial Neural Networks (ICANN 2019), IEEE Transactions of Intelligent Transportation Systems, Springer Artificial Intelligence Review, Springer Scientometrics, Springer Applied Network Science, Elsevier Informetrics